KAVINESH M

8610773275

kavineshm7@gmail.com

https://www.linkedin.com/in/kavinesh-m-718163253

ABOUT ME

Enthusiastic Final year Electronics and Communication Engineering student with a keen interest in Embedded Systems, IoT, Robotics and VLSI. Looking to apply theoretical knowledge to real-time electronics projects and gain industry exposure. Passionate about designing and developing smart, efficient systems. Eager to grow technically and contribute to innovative engineering solutions.

PROJECTS

- **2023-2024**

Simulation in wokwi

IOT BASED ROBOTIC ARM

Simulated an IoT-controlled robotic arm using ESP32 and servo motors on the Wokwi platform. Developed control logic for realistic multi-servo movement and remote operation via the Blynk app. Integrated real-time feedback and validated system functionality through simulation.

- **2024-2025**

Simulation project

AI-POWERED PRECISION IRRIGATION SYSTEM FOR AGRICULTURE

Designed and implemented an AI-powered smart irrigation system for agriculture using ESP32 and Python. Integrated (OpenWeatherMap) for weather forecasting, (Firebase) for cloud monitoring, and Telegram for real-time alerts. Optimized water usage through automated irrigation decisions based on real-time soil moisture and weather data.

- **2024-2025**

3rd year Mini project(Iot)

IOT BASED SMART LOCK SYSTEM

I built a smart door lock system using an ESP32 board. It can be unlocked using an RFID card, a keypad code, or through the Blynk mobile app. The system also includes a buzzer alert, auto-lock after 5 seconds, and shows lock status in real time.

- **2025-2026**

INTERNSHIP at FINSTEIN

ROBOTIC ARM CONTROL USING JOYSTICK

Designed and developed a 3D-printed robotic arm controlled using a joystick and Arduino UNO. Integrated MG995 servo motors for smooth pick-and-place operations. Used an external power adapter to ensure stable motor performance. Gained hands-on experience in embedded systems and servo control.

- **2025-2026**

INTERNSHIP at FINSTEIN

VERTICAL PLOTTER(DRAWING BOT)

During my robotics internship, I built a vertical plotter (glass-wall drawing bot) that automates marker sketches on glass. I designed and integrated the control system using an Arduino Uno with an L293D motor shield, NEMA 17 stepper motors, and a servo for pen lift/drop, and engineered suction-cup mounting for secure operation on vertical surfaces. I modeled and 3D-printed custom mounts and housings, and developed firmware for coordinated stepper motion, steps/mm calibration, and precise stroke sequencing. I validated performance through calibration and multi-sketch testing and documented wiring.

- **2025-2026**

INTERNSHIP at FINSTEIN

SPIDER ROBOT (XBOT)

Designed and built a 3D-printed quadruped robot that delivers a compact, rigid chassis with articulated legs and clean cable routing for dependable leg motion and easy maintenance. Integrated an Arduino Nano with a PCA9685 16-channel PWM driver to coordinate eight MG90S metal-gear micro servos over I2C, enabling precise multi-joint actuation with reduced pin usage and low control jitter. Implemented synchronized gait sequencing to execute crawl and trot patterns, improving low-speed stability and turning responsiveness for consistent ground contact and stride timing. Powered the platform with a lithium-ion battery pack and separated logic/servo power rails to sustain reliable operation during dynamic gaits. Documented wiring, calibration steps, and operating procedures to enable rapid replication and future sensor or teleoperation upgrades

EDUCATION

- **2022-2026**

Pursuing

P.T.LEE.CHENGALVARAY NAICKER COLLEGE OF ENGINEERING AND TECHNOLOGY

- B.E in Electronic and Communication Engineering

- **2021 - 2022**

75%

SKV MATRICULATION SCHOOL

- HSC

- **2019-2020**

66%

SRI RAMANA MAHARSHI MATRICULATION SCHOOL

- SSLC

SKILLS

- Embedded Systems (ESP32, Arduino, Raspberry, Flipper zero)
- IoT Development (Blynk, Firebase, ThingSpeak)
- 3D Printing & 3D Designing.
- AI Tools (ChatGPT, copilot, Claude, gemini, perplexity)
- Simulation Tools (Wokwi, SPICE, tinkercad)
- Programming: Python
- Problem-solving & analytical thinking.

CERTIFICATIONS

- VLSI Course – Internshala (2024)
- Presented a paper at "VIDYUT YANTRA 2K25" Symposium – Alpha College of Engineering **(Won 1st Prize)**
- Presented a paper at "XELFUGETS 2K24" Symposium – Jaya Engineering college (**Participation**)
- Completion Certificate for a two-month Robotics Internship at Finstein

LANGUAGES

- Tamil
- English